TOGHRUL ALIYEV

Data Science | Machine Learning | Applied AI Engineering

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SUMMARY

Data Science student with hands-on experience developing practical machine learning and AI solutions across large-scale data, deep learning, RAG, and end-to-end ML pipelines. Experienced with Python, PyTorch, LLM APIs, retrieval systems, and statistical validation, including a ConvLSTM model achieving $R^2 = 0.94$ on 1 GB+ of DTS sensor data.

EDUCATION

BEU × INHA University — B.S. Data Science & Analytics (Dual Degree) | 2025–2029 | Baku, Azerbaijan | GPA: 93.9
Coursework: Statistics, Data Analysis, Machine Learning, Python Programming, Calculus I & II, Physics I & II, etc.

EXPERIENCE

Machine Learning Competitor | Kaggle | Jul 2026 – Present
- Participating in machine learning competitions, focusing on deep learning, computer vision, and applied AI.
- RSNA 2026 – Knee Abnormality Detection: Best reported Public LB: 0.922 macro ROC-AUC (team ensemble).
- Developed and evaluated DINOv2-based pipelines for detecting 12 abnormalities from knee MRI; explored pseudo-labeling, cross-validation, probability calibration, and community-based ensemble methods.

Machine Learning Engineer — Internship | FlyRank AI | Remote | Aug–Sep 2026
- Engineered machine learning models and data workflows for AI-driven analytics and production integration.

Co-Founder | AEN (Academic Exchange Program) | Baku, Azerbaijan (Hybrid) | Mar 2026 – Present
- Leading community partnership development, cross-functional team coordination, and student academic exchange initiatives.

AI Engineering Intern | Microsoft AI Innovators Summer Program | Remote | May–Jul 2026
- Built end-to-end RAG systems and analytical pipelines integrating LLM APIs, covering ingestion, preprocessing, retrieval, and response generation.
- Engineered Python data-processing and retrieval pipelines and applied statistical validation to evaluate system output quality.
- Delivered multiple working AI projects and stakeholder presentations demonstrating system performance and practical AI engineering workflows.

Section Leader — Mentor | Stanford Code in Place | Remote | Apr–May 2026
- Selected globally as a Section Leader/Mentor; led weekly Python fundamentals and OOP sessions aligned with Stanford CS106A and provided 1-on-1 technical mentoring.

Computer Science Translator — Volunteer | Khan Academy | Remote | May–Aug 2026
- Translated computer science and data-literacy educational content into Azerbaijani as a volunteer contributor, expanding access to technical education.

Research Contributor — ML Systems | Baku Engineering University | Supervised by PhD A. Gayibov | 2025–Jun 2026
- Conducted supervised literature reviews on neural network architectures and meta-heuristic multi-agent decision-making systems; synthesized findings into structured technical analyses published on LinkedIn.

SELECTED PROJECTS

DTS Thermodynamic ML Modeling | Python, PyTorch, Scikit-Learn, Pandas, NumPy, Matplotlib | 2026 | GitHub
- Built an end-to-end ETL and ML pipeline predicting wellbore flow rate from 1 GB+ (~26M lines) of noisy DTS sensor data.
- Engineered dT/dz, d^2T/dz^2, and rolling-statistic features and trained a physics-informed ConvLSTM using AdamW + Cosine Annealing.
- Achieved $R^2 = 0.94$ using a strict 80/20 spatial-depth split designed to prevent data leakage.

ZeroT — AI-Native Cybersecurity Analytics Platform | Next.js 15, React 18, Python, Gemini 2.5 API, Vercel | 2025 | zerot.vercel.app
- Integrated Google Safe Browsing, PhishTank, and URLScan.io feeds into a Gemini 2.5 risk-classification pipeline processing 100+ URLs simultaneously.
- Built a real-time OSINT dashboard with plain-language reporting; reached the GDG Baku Hackathon National Final.
- Implemented an anti-hallucination validation layer supporting sub-second AI risk assessments.

TECHNICAL SKILLS

Languages: Python, C++, SQL (Intermediate), Bash
Data & ML: Pandas, NumPy, Polars, Scikit-Learn, Matplotlib, Seaborn, EDA, Feature Engineering, Statistical Validation
Deep Learning & AI: PyTorch, Hugging Face Transformers, LangChain, LangGraph, RAG, LLM APIs
Data Engineering: ETL, Data Preprocessing, End-to-End ML Pipelines
Tools & Platforms: Databricks, Jupyter, Google Colab, Kaggle, Docker, Git/GitHub, Linux, Vercel, Render, Railway

RESEARCH

The Neural Network Family: A Guide to Every Major Architecture | Deep Learning Architecture Design | Supervised by PhD A. Gayibov, BEU | Published on LinkedIn (2025–Jun 2026)
Covers ANN, FNN, CNN, RNN/LSTM/GRU, Transformer, GNN, and SNN, mapping architectures to their structural rationale and problem classes.

CERTIFICATIONS & COMPETITIONS

Certifications: Machine Learning Specialization — DeepLearning.AI (Andrew Ng) • Data Analysis with Python — IBM/Coursera • Prompting Essentials — Google • Advanced Prompting — Google • Additional certifications available on LinkedIn
Competitions: Build with AI — GDG Hackathon — Finalist (2026) • AI Automation Hackathon (n8n & Fal.ai) (2026) • Kaggle — Active competitive data science; currently participating in an RSNA competition

LANGUAGES

English (Advanced) • Turkish (Advanced) • Azerbaijani (Native) • Korean (Limited Proficiency) • Russian (Limited Proficiency)